

Eta-Quotients in Lean 4

Cusp Orders at Every Cusp, Ligozat's Criterion at Small Level,
and a Named Obstruction at General Level

PREPRINT — not peer reviewed

Xavier Callens*

SocrateAI Research Initiative

September 7, 2026

Archived at Zenodo: [10.5281/zenodo.22648098](https://zenodo.org/record/22648098) (all versions)

Status. This is a *preprint*. It has **not been peer reviewed**. The Lean development it describes compiles and is axiom-audited — those claims are machine-checked and independently reproducible — but the *exposition* has had no external referee. One headline result is deliberately bounded (§4): Ligozat's transformation law is proved for $N \in \{1, 2, 3, 4, 5, 7, 13\}$ and **not** in general, the general case being blocked by the obstruction named in §5. This version incorporates revisions from one external review and one internal adversarial pass (the response is archived alongside the deposit); neither constitutes venue peer review. Read the scope before citing the result.

Abstract

We formalize in Lean 4, on top of Mathlib, the arithmetic and analytic layers of Ligozat's criterion for eta-quotients $f(z) = \prod_{\delta|N} \eta(\delta z)^{r_\delta}$, and we report one theorem, one bounded theorem, and one obstruction. **(i)** For *every* level N , every integer exponent vector r with $\sum_{\delta|N} r_\delta = 2k$, and *every* $\gamma \in SL_2(\mathbb{Z})$ — with no congruence condition and no $\Gamma_0(N)$ -membership hypothesis — we prove the two-sided asymptotic $\|(f|_k \gamma)(z)\| = \Theta(\exp(-2\pi \text{ord}^{\text{raw}}(N, r, c) \text{Im } z))$ along $\text{Im } z \rightarrow \infty$, where $c = \gamma_{10}$ and $\text{ord}^{\text{raw}}(N, r, c) = \frac{1}{24} \sum_{\delta|N} \gcd(c, \delta)^2 r_\delta / \delta$. At the cusp ∞ we additionally identify the order in *Mathlib's own* language, `meromorphicOrderAt (cuspFunction 1 f) 0`. **(ii)** We prove Ligozat's transformation law in full — $f(\gamma z) = w(\gamma)(cz + d)^k f(z)$ on *all* of $\Gamma_0(N)$, with the character identified — for $N \in \{1, 2, 3, 4, 5, 7, 13\}$. **(iii)** For general N we do not prove it, and we explain precisely why: the multiplier is $w(\gamma) = \exp(\frac{\pi i}{12} \text{per}(\gamma))$ where *per* is the period cocycle of the weight-two quasi-modular combination $\sum_{\delta} r_\delta \delta E_2(\delta z)$; for a single η this period function is Rademacher's Φ , whose non-coboundary content is the Dedekind sum $s(d, |c|)$. Mathlib contains no Dedekind sums. We state the minimal Mathlib addition that would remove the obstruction. The development is 519 declarations in 8117 lines across five modules, with no `sorry`, and 391 build-failing axiom guards.

MSC 2020: 11F20, 11F11, 68V20. **Keywords:** eta-quotients, Ligozat's criterion, Dedekind eta, Dedekind sums, Rademacher Φ , Lean 4, Mathlib.

1 Introduction

An *eta-quotient* of level N is a product

$$f(z) = \prod_{\delta|N} \eta(\delta z)^{r_\delta}, \quad r_\delta \in \mathbb{Z},$$

*Corresponding author: callensxavier@gmail.com

over the divisors of N , where η is Dedekind's function. Ligozat's criterion states that f is a modular form on $\Gamma_0(N)$ of weight $k = \frac{1}{2} \sum_{\delta} r_{\delta}$ provided

$$\sum_{\delta|N} \delta r_{\delta} \equiv 0 \pmod{24}, \quad \sum_{\delta|N} \frac{N}{\delta} r_{\delta} \equiv 0 \pmod{24}, \quad (1)$$

and the order of f at every cusp is non-negative [1, 2, 3]. It is the standard working tool for producing modular forms by hand, and the reason eta-quotients appear throughout the theory of partitions and of monstrous moonshine.

This paper reports what happens when one tries to formalize it. The honest answer is: the arithmetic and the analysis go through in general; the *group theory* does not, and the obstruction is sharp and classical.

What Mathlib provides, and what it does not

We are explicit about the substrate because it determines what is new. At commit 905b9581 (2026-07-28) Mathlib contains, in `NumberTheory/ModularForms/`:

- `DedekindEta.lean` — 20 declarations: η as the q -product $\eta(z) = q^{1/24} \prod_n (1 - q^{n+1})$, non-vanishing (`eta_ne_zero`), differentiability, and $\log \text{Deriv } \eta = \frac{\pi i}{12} E_2$;
- `Discriminant.lean` — crucially, the S -transformation of η : `eta_comp_eq_csqrI_inv` gives $\eta(-1/z) = (\sqrt{i})^{-1} \sqrt{z} \eta(z)$, together with the modularity of $\Delta = \eta^{24}$;
- `Cusps.lean` (`IsCusp`, `CuspOrbits`, `widthInfty`), `QExpansion.lean` (`cuspfFunction`, `qExpansion`), and the general order machinery `analyticOrderAt` / `meromorphicOrderAt`.

Repository-wide searches return **zero** occurrences of `EtaQuotient`, `Ligozat`, `DedekindSum`, `dedekindSum` and `etaMultiplier`. The last two absences are the subject of §5.

Remark 1.1 (A methodological warning we paid for). An earlier pass in this project concluded that Mathlib had no order-of-vanishing machinery, on the strength of `grep -rl 'orderAt'` returning zero files. It does have it: the declarations are `analyticOrderAt` and `meromorphicOrderAt`, and `grep -rli 'orderat'` returns nineteen. A zero result from a case-sensitive search on a lower-camelCase name is not evidence of absence. Correcting this turned an assumed obstruction into Theorem 3.3.

2 The arithmetic layer

The combinatorial content is separated into a module mentioning no modular form, no η and no upper half-plane: it is a statement about r and N alone. The central definition is Ligozat's cusp-order expression, for $d \mid N$,

$$\text{ord}(N, r, d) = \frac{N}{24} \sum_{\delta|N} \frac{\gcd(d, \delta)^2 r_{\delta}}{\gcd(d, \frac{N}{d}) d \delta},$$

transcribed from [3, Thm. 1.64], together with the two congruences of (1) as predicates `LigozatCongr1` and `LigozatCongr2`. The divisor involution $\delta \mapsto N/\delta$ (Mathlib's `Nat.sum_div_divisors`) gives the duality exchanging the two congruences, formalized as `ligozatCongr1_dual`.

3 Cusp orders, and a correction to our own specification

Theorem 3.1 (Order at every cusp, general N). *etaQuotient_cusp_order*. For every N , every r with $\sum_{\delta} r_{\delta} = 2k$, and every $\gamma \in SL_2(\mathbb{Z})$, writing $c = \gamma_{10}$,

$$\|(f|_k \gamma)(z)\| = \Theta_{\text{Im } z \rightarrow \infty} \exp(-2\pi \text{ord}^{\text{raw}}(N, r, c) \text{Im } z), \quad \text{ord}^{\text{raw}}(N, r, c) = \frac{1}{24} \sum_{\delta|N} \frac{\gcd(c, \delta)^2 r_{\delta}}{\delta}.$$

```
def etaQuotientRawOrder (N : ℕ) (r : EtaExp) (c : ℤ) : ℚ :=
  (∑ δ ∈ N.divisors, ((Int.gcd c (δ : ℤ) : ℚ) ^ 2 * (r δ : ℚ)) / (δ : ℚ)) / 24

theorem etaQuotient_cusp_order (N : ℕ) (r : EtaExp) {k : ℤ}
  (hk : ∑ δ ∈ N.divisors, r δ = 2 * k) (γ : SL(2, ℤ)) :
  (fun z : ℍ => ||(etaQuotientH N r |k| ((γ : SL(2, ℤ)) : GL (Fin 2) ℝ)) z||)
  =0[UpperHalfPlane.atImInfy]
  fun z : ℍ => Real.exp (-2 * π * ((etaQuotientRawOrder N r (γ 1 0) : ℚ) : ℝ) * z.im) := by
```

Listing 1: The raw cusp order, and Theorem 3.1. The two-sided bound is Mathlib’s `Asymptotics.IsTheta` at the filter `UpperHalfPlane.atImInfy`, written `=0[·]`; no approach region appears because that filter constrains only $\text{Im } z$.

Two features deserve emphasis. There is *no* congruence hypothesis and *no* $\Gamma_0(N)$ -membership hypothesis: the asymptotic holds for every γ in the full modular group. And the Θ -relation is taken along the filter `atImInfy`, which constrains only $\text{Im } z$; the statement is therefore uniform in $\text{Re } z$ and no approach region is assumed.

Remark 3.2 (The width normalisation — our specification was wrong). This node was originally specified with the exponent written as Ligozat’s $\text{ord}(N, r, c)$. *That statement is false*, and it is not what was proved. Ligozat’s $\text{ord}(N, r, d)$ is an order of vanishing in the local uniformiser $q_h = e^{2\pi iz/h}$ at a cusp of denominator d , where $h = N/\gcd(d^2, N)$ is the width of that cusp for $\Gamma_0(N)$. A decay statement in $\text{Im } z$ therefore carries the exponent ord/h , not ord . Formally, `etaQuotientCuspOrder_eq_width_mul_raw` gives $\text{ord}(N, r, d) = h \cdot \text{ord}^{\text{raw}}(N, r, d)$, so the two exponentials differ whenever $h \neq 1$. The companion statement `etaQuotient_cusp_order_ligozat` spells the exponent as ord/h so that the correction is visible in the statement itself.

Ligozat’s condition (iii) is untouched: $h > 0$, so $\text{ord}^{\text{raw}} \geq 0$ at a cusp if and only if $\text{ord} \geq 0$ there. A hand-computed pin is kept in the development: $f = (\eta(z)\eta(2z))^{12}$ satisfies $\|f|_{12}S\| \asymp \exp(-3\pi y/2)$.

Theorem 3.3 (The cusp ∞ , in Mathlib’s own language).

meromorphicOrderAt_cuspFunction_eq_cuspOrder_infy.

For every N , every r , and every $m \in \mathbb{Z}$ with $24m = \sum_{\delta|N} \delta r_{\delta}$,

$$\text{meromorphicOrderAt}(\text{cuspFunction}_1(f))(0) = m,$$

and $m = \text{ord}(N, r, N)$, Ligozat’s value at $d = N$.

This uses Mathlib’s own notion of order, its own cusp function, and its own width (`strictWidthInfy` of $\Gamma_0(N)$ is 1); we introduce no competing notion of “order at a cusp”. There is no positivity and no holomorphy hypothesis — the order may be negative.

4 Ligozat’s transformation law at small level

Theorem 4.1 (Full transformation law, $N \leq 4$). *ligozat_of_le_four*. Let $0 < N \leq 4$, let $\sum_{\delta|N} r_{\delta} = 2k$, and assume both congruences (1). Let $w : \Gamma_0(N) \rightarrow \mathbb{C}^{\times}$ be any homomorphism

with $w(T) = 1$, $w(V) = 1$ and $w(-I) = (-1)^k$, where $V = W_N T W_N^{-1}$. Then for every $\gamma \in \Gamma_0(N)$ and every $z \in \mathbb{H}$,

$$f(\gamma z) = w(\gamma) (cz + d)^k f(z).$$

The architecture is worth stating because it is what makes the bounded result possible. A general construction produces a character with this transformation law for every N and with no congruence conditions; what it cannot do is say *which* character. Ligozat’s first congruence pins it at T ; the second pins it at $V = W_N T W_N^{-1}$ — and W_N here is the Fricke involution from the companion development [8], which is where that work is reused; a separate argument pins it at $-I$. For $N \leq 4$ those three elements generate $\Gamma_0(N)$, by a Euclidean descent, so the character is determined and the congruences are upgraded from “conditions at two generators” to the transformation law on the whole group.

```

noncomputable def etaQuotientModularForm {N : ℕ} (hN : 0 < N) (hN4 : N ≤ 4) (r : EtaExp) {k : ℤ}
  (hk : ∑ δ ∈ N.divisors, r δ = 2 * k) (h1 : LigozatCongr1 N r) (h2 : LigozatCongr2 N r)
  (hke : Even k)
  (hbd : ∀ γ : SL(2, ℤ), IsBoundedAtImInfty ((etaQuotientH N r) |[k] γ)) :
  ModularForm (Gamma0 N) k where
toFun := etaQuotientH N r
slash_action_eq' := etaQuotientH_slash_of_le_four hN hN4 r hk h1 h2 hke
holo' := mdiff_etaQuotientH N r
bdd_at_cusps' hc := isBoundedAt_of_bdd_SL2Z r hbd hc

```

Listing 2: The eta-quotient as a term of Mathlib’s bundled **ModularForm**, for $N \leq 4$ and even k . Ligozat’s condition (iii) enters as the hypothesis **hbd**; see Remark 4.2.

Remark 4.2 (The bundled **ModularForm**, and what condition (iii) costs). For $N \leq 4$ and *even* k the eta-quotient is a term of Mathlib’s bundled **ModularForm (Gamma0 N) k**: **etaQuotientModularForm**, whose three fields are the transformation law of Theorem 4.1, holomorphy, and boundedness at the cusps. So the assembly is done, not outstanding — but it takes Ligozat’s condition (iii) as the hypothesis **hbd**, boundedness of every $SL_2(\mathbb{Z})$ -translate at $i\infty$, rather than deriving it from non-negativity of the cusp orders. Deriving it needs the order of vanishing at a *general* cusp, and that is the obstruction of §5 again: the same missing multiplier. Theorem 3.1 supplies the analytic content at every cusp, so what is missing is the bridge from a Θ -asymptotic to **IsBoundedAtImInfty** at a cusp whose width is not 1 — API, resting on an obstructed input.

The even- k hypothesis is not bookkeeping either. For odd k the conclusion is *false*: at $N = 3$, $r_1 = -3$, $r_3 = 9$ both congruences hold and $\sum r_\delta = 6 = 2 \cdot 3$, yet the trivial-multiplier conclusion fails at $\gamma = -I$. That counterexample is formalized as a guard.

Theorem 4.3 (Genus-zero primes). *ligoat_of_prime*. The same conclusion holds for $p \in \{5, 7, 13\}$.

The descent fails at $N = 5$, and not by accident: the development records the explicit counterexample **descent_bound_fails_at_five**. Classically $\bar{\Gamma}_0(5)$ is the free product $\mathbb{Z}/2 * \mathbb{Z}/2 * \mathbb{Z}$, so $\{-I, T, V\}$ cannot generate it and two elliptic generators are needed. That necessity is a literature fact and is *not* formalized here; what is formalized is that this proof needs them — deleting the two elliptic generators and re-running the block leaves exactly two unprovable goals. The replacement tool is a Schreier transversal criterion, **Gamma0_eq_of_schreier**, valid at every prime p : a subgroup containing $-I, T, V$ equals $\Gamma_0(p)$ as soon as it contains one Schreier generator for each $j \in [1, p - 1]$. We supply those generators at the three primes where $X_0(p)$ has genus zero.

5 The obstruction at general level

We do not prove Ligozat’s transformation law for general N , and the reason is not a shortage of effort or a choice of proof strategy. It is exact.

Write the multiplier as $w(\gamma) = \exp(\frac{\pi i}{12} \text{per}(\gamma))$, where per is the period cocycle of the weight-two quasi-modular combination $\sum_{\delta|N} r_\delta \delta E_2(\delta z)$. This period is precisely the integration constant that both natural soft routes destroy: the `logDeriv` route differentiates it away, and the 24th-power route (passing to f^{24} , a quotient of discriminants whose modularity Mathlib already has) quotients it away. For a single η this period function *is* the classical Dedekind–Rademacher Φ [5, Thm. 3.11]: for $c \neq 0$,

$$\Phi(\gamma) = \frac{a+d}{c} - 12 \text{sign}(c) s(d, |c|),$$

whose non-coboundary content is exactly the Dedekind sum $s(d, |c|)$. Its conjugation-invariant homogenization, the Rademacher symbol $\Psi(\gamma) = \Phi(\gamma) - 3 \text{sign}(c(a+d))$, is what the E_2 -period computes over the closed geodesic of a hyperbolic γ [4]; the multiplier itself uses Φ , and it is Φ that obeys the cocycle law quoted below.

The reason is group-theoretic. A character into μ_{24} factors through the abelianization, and for $\Gamma_0(N)$ — a Fuchsian group of genus $g = \text{genus}(X_0(N))$ with s cusps — the free rank of the abelianization is $2g + s - 1$ (a classical fact, not formalized here). The congruences pin w only at T and at $V = W_N T W_N^{-1}$, hence on a subgroup of the free part of rank at most two, and torsion relations pin it at elliptic points; on a generator of infinite order, a relation such as $w^{24} = 1$ says only that the image lies in μ_{24} , never *which* root of unity it is. The small-level successes of §4 are exactly the levels where this bookkeeping closes: for $N \leq 4$ the free rank is at most two, and at $p \in \{5, 7, 13\}$ it is one, the elliptic pins supplying the rest. The first failure is parabolic, not hyperbolic: $\Gamma_0(6)$ is torsion-free of free rank three (genus zero, four cusps), so $\{-I, T, V\}$ cannot generate it, and the extra parabolic classes could only be pinned through q -expansions at the extra cusps — precisely the general-cusp machinery this section shows to be obstructed. From genus one on (for instance $N = 11$, and most $N \geq 17$) the situation is strictly worse: $2g$ independent homology classes are represented by *hyperbolic* elements, on which no congruence, torsion relation, or cusp expansion says anything, and pinning w there requires evaluating the period integral explicitly — which is what forces Dedekind sums into the argument. Evaluating w in closed form on a single hyperbolic element of $\Gamma_0(11)$ is mathematically equivalent to evaluating a Dedekind sum. Every disguise we attempted reduces back to this: Atkin–Lehner conjugation reintroduces general eta transforms, theta/Poisson routes re-derive the same 24th-root phases, and Wohlfahrt level-24 arguments presuppose the multiplier formula.

What would remove it

In ascending strength, the minimal Mathlib additions are:

- (1) the Dedekind sum $s(d, c)$ and reciprocity;
- (2) Rademacher’s $\Phi : SL_2(\mathbb{Z}) \rightarrow \mathbb{Z}$ with the cocycle law $\Phi(AB) = \Phi(A) + \Phi(B) - 3 \text{sign}(c_{ACBCAB})$;
- (3) the Petersson–Rademacher multiplier $\eta(\gamma z) = \exp(\pi i (\frac{a+d}{12c} - s(d, c) - \frac{1}{4})) \sqrt{cz+d} \eta(z)$ for $c > 0$.

With (3), general- N Ligozat becomes moderate bookkeeping on top of the argument already formalized here. We note that (2) is best constructed as the period of E_2 using the `E2_slash_action` machinery *already present* in Mathlib — the analytic half exists — which makes it the cleanest available upstream contribution.

Remark 5.1 (A per-instance escape hatch). For a *fixed* N and r , the multiplier on a finite generating set is a ratio of convergent q -products valued in the discrete set μ_{24} . Interval arithmetic with error below $|1 - \zeta_{24}|/2$ therefore *proves* each value. This is an *A2* (program-checked) route for any concrete eta-quotient at any level, requiring no Mathlib addition, though heavy to promote to kernel-checked *A1*.

6 Verification status

lake build SocrateAI completes in 3462 jobs with no errors. The development is 519 declarations in 8117 lines across five modules (EtaQuotient, EtaQuotientCuspOrder, EtaQuotientCuspTheta, EtaQuotientModularity, EtaQuotientPrimeLevel). There are no sorry placeholders, and the five eta modules declare no axioms. (One legacy module elsewhere in the library, Generated/BlueprintSkeleton.lean, declares five vacuous axiom ... : True postulates from an early scaffold; they are imported nowhere, and the enforced footprints below prove that no guarded theorem depends on them.) Following [7], the axiom footprint is enforced rather than reported: FinalCheck.lean carries 391 #guard_msgs in #print axioms guards over 389 distinct theorems, so a widened footprint is a compile error; 385 report exactly [propext, Classical.choice, Quot.sound], one the strictly smaller [propext, Quot.sound], and three (definitional guards) depend on no axiom at all. An earlier version of this paragraph said 370; that number was an artifact of counting the info strings with a single-line search, which misses the ones that wrap. A negative control asserting a deliberately wrong footprint is kept outside the build and verified to fail, which is what establishes the guards are load-bearing rather than vacuous.

Of the 453 theorems in the new modules, 14 (3%) have a one-line rfl/decide proof, and 45% quantify over a structure. We report this ratio because an earlier module in this project consisted of 174 numeral identities of which 158 were one-line, and carried no evidential weight; the check is now standard practice here for any large batch.

The statement-dependency graph (dag/theorems.jsonl, 57 nodes, 51 proved, acyclic) is machine-validated: every node marked proved must name a declaration that resolves and may depend only on proved nodes. **The node ETA-01, Ligozat’s criterion in general, remains open with lean_name: null**, and the obstruction node is blocked with lean_name: null — for that node the absence of a declaration *is* the content.

Artifact. Repository <https://github.com/xaviercallens/SocrateAI-Lean-Lib>; the five modules EtaQuotient*.lean under Lean/SocrateAI/ModularForms/, axiom guards in Lean/SocrateAI/FinalCheck.lean, dependency graph in dag/. Toolchain leanprover/lean4:v4.32.2; Mathlib pinned at 905b95818eb32af7874a58b427f50c1711a5e96c (2026-07-28), which is the revision every theorem here was checked against. The build configuration is portable — lakefile.lean requires Mathlib from git at that revision and no tracked file contains a machine-specific path — so git clone, lake exe cache get, lake build SocrateAI reproduces the artifact; see BUILDING.md. Cite this preprint through its concept DOI [11], which resolves to the newest version. Mirrored at <https://huggingface.co/datasets/callensxavier/socrateai-eta-quotients>.

7 Related work and prior art

Mathlib has no eta-quotients. The Fermat’s Last Theorem artifact [7] contains substantial eta-related material, but a direct reading shows it is about modular *units* on modular function fields — modularUnitSeries, sharpUnit*, over Laurent and Hahn series. Its three occurrences of “Ligozat” are *namespace labels* (LigozatUnitEngine, LigozatUnitAL) on that machinery, not a formalization of the modularity criterion. A GitHub-wide language:lean search returns zero

for **Ligozat**. We did *not* search the Lean Zulip or open mathlib4 pull requests, and we do not claim coverage there.

Other proof assistants. The component we identify as obstructed has been mechanized elsewhere, which is evidence that the obstruction is a gap in Mathlib rather than a difficulty of principle. The Isabelle/HOL Archive of Formal Proofs contains *Dedekind Sums* [9], which defines $s(h, k) = \sum_{r=1}^{k-1} \frac{r}{k} \left(\left\{ \frac{hr}{k} \right\} - \frac{1}{2} \right)$ and proves reciprocity, and *Rademacher's Series for the Partition Function* [10], which uses that machinery in exactly the analytic setting where it arises here. Our §5 therefore asks Mathlib for something Isabelle/HOL already has; the constructions there are a usable guide, and the existence of two independent formalizations of Dedekind sums is a reason to expect the Mathlib addition to be routine rather than research.

8 Limitations

Remark 8.1 (What is not done).

- (a) Ligozat's criterion for general N is not proved (§5).
- (b) The order of vanishing at a cusp *other than* ∞ is not stated in Mathlib's `meromorphicOrderAt` language. Mathlib's width is `widthInfty` only; there is no `widthAtCusp` and no `orderAtCusp`. These are constructible from `conjGL`, `widthInfty` and the slash action, so this is missing API rather than an obstruction. Theorem 3.1 gives the analytic content at every cusp regardless.
- (c) The indexing data is not formalized: that the cusps of $\Gamma_0(N)$ are parametrised by $d \mid N$ with multiplicity $\varphi(\gcd(d, N/d))$, and that $[SL_2(\mathbb{Z}) : \Gamma_0(N)] = \psi(N)$. No index formula for Γ_0 exists anywhere in Mathlib. The definitions `numCusps` and `dedekindPsi` in this development carry those classical readings but do not prove them.
- (d) Ligozat's condition (iii) is not *derived* from the cusp-order formula. The bundled `ModularForm` exists (Remark 4.2) but takes boundedness as a hypothesis; discharging it from non-negativity of the cusp orders needs the order at a general cusp, which is (b), which is the obstruction.
- (e) That $\{-I, T, V\}$ *cannot* generate $\Gamma_0(5)$ is cited, not proved.

References

- [1] G. Ligozat, *Courbes modulaires de genre 1*, Bull. Soc. Math. France, Mém. **43** (1975).
- [2] Y. Martin, *Multiplicative eta-quotients*, Trans. Amer. Math. Soc. **348** (1996), 4825–4856.
- [3] K. Ono, *The Web of Modularity*, CBMS Regional Conference Series in Mathematics **102**, AMS (2004). doi:10.1090/cbms/102.
- [4] H. Rademacher, *Topics in Analytic Number Theory*, Grundlehren **169**, Springer (1973).
- [5] T. M. Apostol, *Modular Functions and Dirichlet Series in Number Theory*, 2nd ed., GTM **41**, Springer (1990).
- [6] The mathlib Community, *The Lean mathematical library*, CPP 2020.
- [7] Anthropic, *A formal proof of Fermat's Last Theorem in Lean 4*, software artifact, <https://github.com/anthropics/fermats-last-theorem>, 2026.

- [8] X. Callens, *A Mathlib-Idiomatic Formalization of the Fricke Involution and its Operator on Modular Forms*, Zenodo, 2026. doi:10.5281/zenodo.22542571.
- [9] M. Eberl, A. Bordg, L. C. Paulson, W. Li, *Dedekind Sums*, Archive of Formal Proofs, 2025. https://www.isa-afp.org/entries/Dedekind_Sums.html, formal proof development.
- [10] M. Eberl, L. C. Paulson, *Rademacher's Series for the Partition Function*, Archive of Formal Proofs. https://www.isa-afp.org/entries/Rademacher_Series.html, formal proof development.
- [11] X. Callens, *Eta-Quotients in Lean 4* (this preprint), Zenodo, 2026. doi:10.5281/zenodo.22648098.
- [12] L. de Moura et al., *The Lean 4 Theorem Prover and Programming Language*, CADE-28, LNCS 12699, Springer (2021).